

Payton Pei

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SUMMARY

Systems Science and Engineering master's student at the University of Ottawa, graduating in December 2026 (expected). Software engineer building and fine-tuning AI systems with Python, PyTorch, Rust/Tauri and TypeScript/React; published LifeMint on the App Store. Former Nokia and Ford intern seeking new-grad AI application and developer-tools roles in Hangzhou.

EDUCATION

University of Ottawa

Master of Systems Science and Engineering

2024 — Dec 2026 (expected)

Carleton University

Bachelor of Computer Science

2017 — 2023

PROJECTS

GaokaoTutor — LLM Fine-Tuning & Evaluation

github.com/mammut001/gaokao

- Fine-tuned Spark X2.5-4B with 4-bit QLoRA on an RTX 4090 using 161 tutoring traces; built reproducible data provenance, anti-contamination checks and a matched evaluation on a frozen 80-case unseen holdout.
- Reduced unsolicited solution dumping in hint/diagnosis cases from 100% to 13.5% and truncation from 57.5% to 0%; integrated the adapter into a Python tutor service and Expo prototype with deterministic math verification.

pipi-shrimp-agent

github.com/mammut001/pipi-shrimp-agent

- Built a local-first desktop AI agent with Rust/Tauri, React/TypeScript and SQLite, integrating multiple LLM providers, structured tool calls and project memory.
- Implemented provider adapters, buffered SSE parsing and cancellation, plus CDP browser automation with failure recovery and context compression for long-running sessions.

LifeMint (formerly Focus Mint)

apps.apple.com/us/app/lifemint/id6759029810

- Published a SwiftUI productivity app for iPhone, iPad and Apple Watch with focus sessions, earnings tracking and English, Chinese and French localization.
- Implemented StoreKit purchases, widgets, Live Activities and bidirectional iPhone–Apple Watch synchronization with WatchConnectivity.

WORK EXPERIENCE

Teaching Assistant

Carleton University

2021

Ottawa, Canada

- Teaching assistant for Python 3 (Jan–Apr 2021) and JOUR 4401 (Sep–Dec 2021), covering relational databases, web crawling and data analysis.

Software Tester Intern

Nokia Canada

Sep 2020 — Aug 2021

Ottawa, Canada

- Tested the legacy NFM-P client in the 7750 Service Router context using JavaScript, XML and the Mocha testing framework.
- Documented system architecture and data flows in Confluence.

Software Engineer Intern

Ford Motor Canada

May 2020 — Aug 2020

Ottawa, Canada

- Contributed to in-vehicle and cloud connectivity software in C++11 on QNX/Linux.
- Wrote GoogleTest unit tests, fixed bugs and maintained technical documentation; used Jenkins in the CI/CD workflow.

RESEARCH

Low-Light RAW Image Restoration with Bridge U-Net and DDBM

2026

- Responsible for the initial DDBM implementation in a collaborative low-light RAW image restoration study using SID and ELD data.
- The project evaluated a DDBM-Feature variant at 26.12 dB PSNR and 0.807 MS-SSIM on 598 SID Sony test images.

SKILLS

- **Languages:** Python, TypeScript, JavaScript, Rust, Swift, C++
- **Frameworks:** React, Node.js, Tauri 2, SwiftUI
- **AI & Systems:** PyTorch, Transformers, PEFT/TRL, QLoRA, MLX, LLM APIs, Tool Calling, MCP, SQLite
- **Testing & Delivery:** PyTest, Vitest, Playwright, GoogleTest, Jenkins, Docker
- **Additional:** CDP Browser Automation, WatchConnectivity, StoreKit 2